Derek Sorensen, Ph.D.

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Education

2019 – 2023	**Ph.D. Computer Science** University of Cambridge. Thesis title: *Formal Tools for Specifying Financial Smart Contracts.*
2016 – 2017	**M.Sc. Mathematics and Foundations of Computer Science** University of Oxford. Thesis title: *On the Tautological Ring of* $\mathcal{g}(S^d \times S^d)$.
2003 – 2006	**B.Sc. Mathematics** Brigham Young University.

Employment History

November 2025 – Present — **Researcher**, *Ethereum Foundation*, Remote.
- Formally specifying and verifying core cryptographic primitives in Lean relevant to Protocol Snarkification.
- Researching formal verification, formalized mathematics, cryptography, zero-knowledge proof systems, and verified computation.

Jun 2025 – Present — **Formal Verification Engineer**, *Just Games*, Remote.
- Smart contract design, engineering, and formal verification
- Mathematical and formal proof of DeFi primitives underpinning the expected value of wagers on uncertain events.
- Technical writing, including the white paper and other formal documentation.

Jan 2024 – April 2025 — **Formal Verification Wizard**, *Certora*, Cambridge, UK.
- Formally verified top-tier EVM DeFi protocols like Silo and Safe.
- Developed new DeFi security products.
- Managed client relationships and project deliverables.

Oct 2022 – Mar 2023 — **Software Engineer**, *Protocol Labs*, Remote.
- Contributor to Lurk, a LISP dialect for zero-knowledge proofs.
- Contributed to specification of meta-Lurk, enabling recursive reference to and computation over zero-knowledge proofs.

May 2021 – Aug 2022 — **Business Development: Sustainability**, *TriliTech*, London, UK.
- Advised businesses in sustainability building on Tezos, including in tokenized carbon offsets, regenerative finance (ReFi), and art marketplaces.
- Member of the Technical Advisory Committee, the ecosystem funding body for the Tezos Foundation.

Mar 2021 – May 2021 — **Consultant: Decentralized Finance**, *Gro*, London, UK.
- Advised on decentralized deposit insurance on the Ethereum Blockchain.
- Built optimization model to fine-tune protocol design and parameters.

Jan 2021 – Mar 2021 — **Consultant: Intellectual Property on Blockchains**, Cambridge, UK.
- Identified industries most likely to tokenize intellectual property rights.
- Delivered technical report on tokenization of intellectual property rights and how it integrates with existing business models and regulation.

Oct 2019 – Nov 2020 — **Formal Verification Engineer**, *Clearmatics*, London, UK.
- Designed, formally specified, and formally verified enterprise blockchains.
- Advised the software development life cycle to formally verify EVM contracts.
- Gave lectures and seminars to the engineering team on formal methods.

Employment History (continued)

Jul 2019 – Sep 2019 **Consultant: Cryptocurrency Design**, *Digital Asset*, New York, NY.
- Designed and built a functional cryptocurrency that supports legal compliance and business-friendly privacy features.
- Specified privacy model, fee and minting structure, and incentive systems.

Mar 2018 – Oct 2018 **Consultant: Blockchain Language Verification**, *RChain Coop*, Seattle, WA.
- Developed the formal semantics of Rholang in the K Framework.

Mar 2018 – Jun 2019 **Research Mathematician**, *Pyrofex Corporation*, Provo, UT.
- Research in the consensus algorithms supporting cryptocurrencies.
- Co-developed the novel and performant consensus algorithm *Casanova*.
- Built the formal semantics for Rholang in K-Framework for the RChain Coop.

2017 – 2019 **Adjunct Faculty (Mathematics)**, *Utah Valley University*, Orem, UT.
- Wrote lecture notes, homework, quizzes, exams.
- Marked and gave feedback to all written work.

Research Publications

1. D. Sorensen, "Formally Specifying Contract Optimizations with Bisimulations in Coq," in *6th International Workshop on Formal Methods for Blockchains (FMBC 2025)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2025, pp. 11–1.
2. S. Jaffer, M. W. Dales, P. Ferris, *et al.*, "Global, robust and comparable digital carbon assets," in *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, IEEE, 2024, pp. 305–306.
3. D. Sorensen, "(In) Correct Smart Contract Specifications," in *2024 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, IEEE, 2024, pp. 567–575.
4. D. Sorensen, "Towards Formally Specifying and Verifying Smart Contract Upgrades in Coq," in *5th International Workshop on Formal Methods for Blockchains (FMBC 2024)*, Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2024, pp. 7–1.
5. D. Sorensen, "Formal Tools for Specifying Financial Smart Contracts," Ph.D. dissertation, 2023.
6. D. Sorensen, "Structured pools for tokenized carbon credits," in *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*, IEEE, 2023, pp. 1–6.
7. D. Sorensen, "Tokenized Carbon Credits," *Ledger*, vol. 8, 2023.
8. K. Butt and D. Sorensen, "Streamlining classical consensus," *International Journal of Blockchains and Cryptocurrencies*, vol. 1, no. 4, pp. 313–328, 2020.
9. D. Sorensen, "Establishing standards for consensus on blockchains," in *International Conference on Blockchain*, Springer, 2019, pp. 18–33.
10. A. Francis, D. Smith, D. Sorensen, and B. Webb, "Extensions and applications of equitable decompositions for graphs with symmetries," *Linear Algebra and its Applications*, vol. 532, pp. 432–462, 2017.

Presentations

2025 **Formally Specifying Contract Optimizations With Bisimulations in Coq. FMBC 2025**, *Hamilton, Canada.*

2024 **(In)Correct Smart Contract Specifications. ICBC**, *Dublin, Ireland.*

Presentations (continued)

	Towards Formally Specifying and Verifying Smart Contract Upgrades in Coq. FMBC, *Luxembourg.*
	Abstraction, Specification, and Proof. Undone Science, *Nantes, FR.*
2023	**Structured Pools for Tokenized Carbon Credits, IEE ICBC CryptoEx**, *Dubai, UAE*
2022	**Applications of Blockchains to Decentralised Finance, Markets, Art, and Beyond, University of Cambridge**, *Cambridge, England*
2021	**Blockchain & DeFi: The Technology and its Applications, Judge Business School, University of Cambridge**, *Cambridge, England*
2019	**Establishing Standards for Consensus on Blockchains. ICBC 2019**, *San Diego.*
2018	**Rholang Semantics and the K-Framework, RCon3**, *Berlin, Germany*
	Formal Verification Panel, RCon3, *Berlin, Germany*
2015	**Presentation at the 2015 BYU Spring Research Conference** Voted best of session
2014	**Presentation at the 2014 BYU Spring Research Conference**

Teaching

Mich. 2020	**Economics, Law, and Ethics**, *University of Cambridge*
	Analysis & Topology, *University of Cambridge*
Summer 2020	**Artificial Intelligence in Blockchain Security**, *Immerse Education*
	Relating the Price of Bitcoin and its Related Cryptocurrencies to Their Respective Trading Volumes, *Horizon Inspires*
	Predicting NBA Playoff Results With Machine Learning, *Horizon Inspires*
Lent 2020	**Groups, Rings, and Modules**, *University of Cambridge*
Spring 2019	**Stat 1040 - Introduction to Statistics**, *Utah Valley University*
Spring 2018	**Stat 1040 - Introduction to Statistics**, *Utah Valley University*
	Stat 2040 - Introduction to Statistics, *Utah Valley University*
	Math 1050 - College Algebra, *Utah Valley University*

Awards

2016	**Robert K Thomas Honors Scholarship**
2015	**AMS Math in Moscow Travel Grant**
	Marc Burton Scholarship
	Best of session at the 2015 BYU Spring Research Conference
2014 & 2015	**Award for Excellence in Undergraduate Research**
	Award for Academic Excellence in Mathematics

Skills

Coding	Rocq, Lean, Solidity, CVL, K Framework
Languages	English (native), Spanish (fluent), Russian (intermediate), French (basic)